

Data Mining Numerical Problems & Solutions

CT 72502 (Elective I) · BEI / BEX IV-I · TU, IOE · Chapter 3: Classification

Ch 3 – Classification – Numerical Problems & Solutions

6 distinct methods · 12 PYQ instances solved · every answer independently recomputed. Decision-tree and Naive-Bayes figures cross-checked against Han & Kamber 3rd ed Examples 8.1–8.4, which use the same dataset the IOE papers copy.

Methods

– M1 Decision tree by information gain (ID3)	79 Bh, 78 Bh, 76 Ch
– M2 Gini index	75 Ch, 71 Ch
– M3 Naive Bayes classification	72 Ch, 79 Bh
– M4 k -Nearest Neighbour	80 Bh
– M5 Confusion matrix metrics	8 papers
– M6 ANN gradients (backpropagation)	76 Ch

M1. Decision tree by information gain (ID3)

Method

1. Compute $Info(D) = -\sum p_i \log_2 p_i$ for the whole set.
 2. For each attribute A : split D , compute $Info_A(D) = \sum \frac{|D_j|}{|D|} Info(D_j)$, then $Gain(A) = Info(D) - Info_A(D)$.
 3. **Highest gain wins** → becomes the node's test.
 4. Recurse on each branch. Stop when a branch is **pure** ($Info = 0$).
- **Time-savers**: a pure partition contributes 0; a 50/50 partition contributes exactly 1; two partitions with the same class ratio have the same $Info$, so compute once.
 - $\log_2 x = \ln x / \ln 2$. Keep 4 decimals until the end.

Problem 1.1 10

79 Bh “Draw decision tree for the given data using ID3 algorithm.” 14 tuples, attributes Age / Income / Student / Credit_Rating, class Buy's_Computer.

Given

14 tuples. Class counts: **Yes = 9, No = 5**.

Step 1: entropy of the whole set

$$Info(D) = -\frac{9}{14} \log_2 \frac{9}{14} - \frac{5}{14} \log_2 \frac{5}{14} = 0.4098 + 0.5305 = \mathbf{0.940}$$

Step 2: gain of each attribute

Attribute	Partition (Yes, No)	$Info_A(D)$	Gain
Age	Youth (2,3) · Middle_Aged (4,0) · Senior (3,2)	$\frac{5}{14}(0.971) + \frac{4}{14}(0) + \frac{5}{14}(0.971) = 0.694$	0.246
Income	High (2,2) · Medium (4,2) · Low (3,1)	$\frac{4}{14}(1.0) + \frac{6}{14}(0.918) + \frac{4}{14}(0.811) = 0.911$	0.029
Student	Yes (6,1) · No (3,4)	$\frac{7}{14}(0.592) + \frac{7}{14}(0.985) = 0.788$	0.151
Credit_Rating	Fair (6,2) · Excellent (3,3)	$\frac{8}{14}(0.811) + \frac{6}{14}(1.0) = 0.892$	0.048

- **Age has the highest gain** → **Age is the root**.
- Note Middle_Aged is (4, 0): **pure**, so that branch is immediately a leaf “Yes”. That purity is what makes Age win.

Step 3: recurse on the Youth branch

(5 tuples: 2 Yes, 3 No, $Info = 0.971$)

- Student = Yes → (2 Yes, 0 No) Student = No → (0 Yes, 3 No)
- Both pure → $Info_{Student} = 0$ → $Gain = 0.971$, the maximum possible.

- **Student** is the test under Youth.

Step 4: recurse on the Senior branch

(5 tuples: 3 Yes, 2 No, $Info = 0.971$)

- Credit = Fair $\rightarrow$ (3 Yes, 0 No) Credit = Excellent $\rightarrow$ (0 Yes, 2 No)
- Both pure $\rightarrow Gain = 0.971$. **Credit_Rating** is the test under Senior.

Final tree

- **Age?**
 - Youth $\rightarrow$ **Student?**
 - no $\rightarrow$ **No**
 - yes $\rightarrow$ **Yes**
 - Middle_Aged $\rightarrow$ **Yes**
 - Senior $\rightarrow$ **Credit_Rating?**
 - fair $\rightarrow$ **Yes**
 - excellent $\rightarrow$ **No**
- Tree has **5 leaves**, depth 2, and classifies all 14 training tuples correctly.

8.2 Decision Tree Induction 331

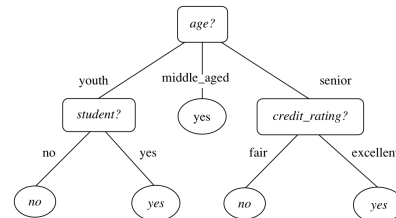


Figure 8.2 A decision tree for the concept *buys_computer*, indicating whether an *AlIElectronics* customer is likely to purchase a computer. Each internal (nonleaf) node represents a test on

The same tree as printed in Han & Kamber 3rd ed, Fig 8.2. Our independently computed gains match their Example 8.1 exactly.

Problem 1.2 8

78 Bh “Construct a decision tree for the following data set using information gain. Predict the class label for <Female, 2, standard, high>.” 10 tuples: Gender / Car ownership / Travel cost / Income level $\rightarrow$ Transport mode.

Given

Classes: Bus = 4, Car = 3, Train = 3. $n = 10$.

$$Info(D) = -\frac{4}{10} \log_2 \frac{4}{10} - \frac{3}{10} \log_2 \frac{3}{10} - \frac{3}{10} \log_2 \frac{3}{10} = 0.5288 + 0.5211 + 0.5211 = \mathbf{1.571}$$

Travel cost

Cheap (4 Bus, 0 Car, 1 Train) · Expensive (0,3,0) · Standard (0,0,2)

- $Info(\text{Cheap}) = -\frac{4}{5} \log_2 \frac{4}{5} - \frac{1}{5} \log_2 \frac{1}{5} = 0.722$
- Expensive and Standard are both **pure** $\rightarrow Info = 0$
- $Info_{TravelCost}(D) = \frac{5}{10}(0.722) + \frac{3}{10}(0) + \frac{2}{10}(0) = 0.361$
- $Gain = 1.571 - 0.361 = \mathbf{1.210}$
- Every other attribute scores far lower (Gender 0.125, Car ownership 0.534, Income level 0.971), so **Travel cost is the root** – two of its three branches are pure.
- Only the *Cheap* branch needs splitting: 5 tuples (4 Bus, 1 Train). Splitting it on **Gender** gives Male (3 Bus, 0 Train) and Female (1 Bus, 1 Train); splitting on **Car ownership** gives 0 (2 Bus), 1 (2 Bus, 1 Train). Neither is perfect; Gender has the higher gain, then Car ownership finishes the Female sub-branch.

Prediction for <Female, 2, standard, high>

- Travel cost = **standard** $\rightarrow$ that branch is **pure Train**.
- **No further tests are needed.** **Answer: Train.**

M2. Gini index

Method

$$Gini(D) = 1 - \sum_i p_i^2 \quad Gini_A(D) = \frac{|D_1|}{|D|} Gini(D_1) + \frac{|D_2|}{|D|} Gini(D_2) \quad \Delta Gini = Gini(D) - Gini_A(D)$$

- **Gini uses BINARY splits only.** For an attribute with v values you must try all $2^{v-1} - 1$ ways of splitting the values into 2 subsets and keep the best.
- Pick the largest $\Delta Gini$ (equivalently the smallest $Gini_A$).

Worked example (same 14-tuple dataset) 6

75 Ch, 71 Ch ask for Gini "with suitable example" – use this. verified/added

- $Gini(D) = 1 - \left(\frac{9}{14}\right)^2 - \left(\frac{5}{14}\right)^2 = 1 - 0.4133 - 0.1276 = \mathbf{0.459}$
- Split **Income** as {low, medium} vs {high}:
 - $D_1 = 10$ tuples (7 Yes, 3 No): $Gini = 1 - (0.7)^2 - (0.3)^2 = 0.420$
 - $D_2 = 4$ tuples (2 Yes, 2 No): $Gini = 1 - (0.5)^2 - (0.5)^2 = 0.500$
 - $Gini_{Income}(D) = \frac{10}{14}(0.420) + \frac{4}{14}(0.500) = 0.300 + 0.143 = 0.443$
 - $\Delta Gini = 0.459 - 0.443 = \mathbf{0.016}$
- Split **Age** as {youth, senior} vs {middle_aged}:
 - $D_1 = 10$ tuples (5 Yes, 5 No): $Gini = 0.500$
 - $D_2 = 4$ tuples (4 Yes, 0 No): $Gini = \mathbf{0}$ (pure)
 - $Gini_{Age}(D) = \frac{10}{14}(0.500) + \frac{4}{14}(0) = 0.357$
 - $\Delta Gini = 0.459 - 0.357 = \mathbf{0.102} \rightarrow \text{best, so Age is chosen}$
- Same winner as information gain, which is the usual (not guaranteed) outcome.

M3. Naive Bayes classification

Method

1. Priors: $P(C_i) = |C_{i,D}|/|D|$.
 2. For each attribute value in X : $P(x_k|C_i)$ from counts within class C_i .
 3. $P(X|C_i) = \prod_k P(x_k|C_i)$.
 4. Compare $P(X|C_i)P(C_i)$ across classes; **largest wins**.
- **Never divide by $P(X)$** – it is identical for every class and cancels.
 - Any zero count $\rightarrow$ apply **Laplacian correction** $\frac{n_k+1}{|C_{i,D}|+k}$.
 - Lay the work out as a small table of conditional probabilities; it earns method marks even if arithmetic slips.

Problem 3.1 10

72 Ch Predict for $X = (\text{age} = \text{youth}, \text{income} = \text{medium}, \text{student} = \text{yes}, \text{credit-rating} = \text{fair})$.

Given

14 tuples. $P(\text{Yes}) = 9/14 = 0.643$, $P(\text{No}) = 5/14 = 0.357$.

Conditional probabilities

	Yes	No
$P(\text{age} = \text{youth})$	$2/9 = 0.222$	$3/5 = 0.600$
$P(\text{income} = \text{med})$	$4/9 = 0.444$	$2/5 = 0.400$
$P(\text{student} = \text{yes})$	$6/9 = 0.667$	$1/5 = 0.200$
$P(\text{credit} = \text{fair})$	$6/9 = 0.667$	$2/5 = 0.400$

Combine

- $P(X|\text{Yes}) = 0.222 \times 0.444 \times 0.667 \times 0.667 = \mathbf{0.0439}$
- $P(X|\text{No}) = 0.600 \times 0.400 \times 0.200 \times 0.400 = \mathbf{0.0192}$
- $P(X|\text{Yes})P(\text{Yes}) = 0.0439 \times 0.643 = \mathbf{0.028}$
- $P(X|\text{No})P(\text{No}) = 0.0192 \times 0.357 = \mathbf{0.007}$

Final answer

$0.028 > 0.007 \rightarrow \text{buys_computer} = \text{Yes}.$

- Matches Han & Kamber 3rd ed Example 8.4.

Problem 3.2 6

79 Bh Classify $X = (\text{Home Owner} = \text{No}, \text{Marital Status} = \text{Married}, \text{Annual Income} = \$120\text{K})$. 10 tuples, class = Defaulted Borrower.

Given

Defaulted: **No = 7, Yes = 3**. $P(\text{No}) = 0.7$, $P(\text{Yes}) = 0.3$.

Step 1: categorical attributes

- $P(\text{HO} = \text{No} | \text{No}) = 4/7 = 0.571$ $P(\text{HO} = \text{No} | \text{Yes}) = 3/3 = 1.0$
- $P(\text{Married} | \text{No}) = 4/7 = 0.571$ $P(\text{Married} | \text{Yes}) = 0/3 = 0 \leftarrow \text{zero!}$
- Apply **Laplacian correction** (Marital Status has $k = 3$ values): $P(\text{Married} | \text{Yes}) = \frac{0+1}{3+3} = 0.167$; and $P(\text{HO} = \text{No} | \text{Yes}) = \frac{3+1}{3+2} = 0.8$.

Step 2: continuous attribute (Income), Gaussian

$$g(x, \mu, \sigma) = \frac{1}{\sqrt{2\pi} \sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

- Class **No**: incomes 125, 100, 70, 120, 60, 220, 75 $\rightarrow \mu = 110, \sigma = 54.54$

$$g(120) = \frac{1}{\sqrt{2\pi}(54.54)} e^{-\frac{(10)^2}{2(2975)}} = 0.00719$$

- Class **Yes**: incomes 95, 85, 90 $\rightarrow \mu = 90, \sigma = 5$

$$g(120) = \frac{1}{\sqrt{2\pi}(5)} e^{-\frac{(30)^2}{2(25)}} = 0.0798 e^{-18} \approx 1.2 \times 10^{-9}$$

- **The Yes class has a tiny spread ($\sigma = 5$) around \$90K, so \$120K is 6 standard deviations out** – effectively impossible under that model.

Step 3: combine

- $P(X|\text{No})P(\text{No}) = 0.571 \times 0.571 \times 0.00719 \times 0.7 = \mathbf{1.64 \times 10^{-3}}$
- $P(X|\text{Yes})P(\text{Yes}) = 0.8 \times 0.167 \times 1.2 \times 10^{-9} \times 0.3 \approx \mathbf{4.8 \times 10^{-11}}$

Final answer

Defaulted Borrower = No (larger by roughly 10^7).

M4. k -Nearest Neighbour**Method**

1. Compute the Euclidean distance from the query to **every** training tuple.
 2. Sort ascending, take the k smallest.
 3. **Majority class** among those k is the prediction.
- **Show the full distance table** – that is where the marks are, not the final label.
 - You may compare squared distances to save time, but write the roots in the final table.

Problem 4.1 6

80 Bh Classify the flower (SepalLength = 7, SepalWidth = 3.2, PetalLength = 4.7, PetalWidth = 1.4), $K = 3$, Euclidean.

Id	Tuple	Squared differences	d	Label
1	(5.1, 3.5, 1.4, 0.2)	$3.61 + 0.09 + 10.89 + 1.44 = 16.03$	4.004	setosa
2	(4.9, 3.0, 1.4, 0.2)	$4.41 + 0.04 + 10.89 + 1.44 = 16.78$	4.096	setosa
3	(4.7, 3.2, 1.3, 0.2)	$5.29 + 0 + 11.56 + 1.44 = 18.29$	4.277	setosa
4	(6.0, 2.2, 4.0, 1.0)	$1 + 1 + 0.49 + 0.16 = 2.65$	1.628	versicolor
5	(6.1, 2.9, 4.7, 1.4)	$0.81 + 0.09 + 0 + 0 = 0.90$	0.949	versicolor
6	(5.6, 2.9, 3.6, 1.3)	$1.96 + 0.09 + 1.21 + 0.01 = 3.27$	1.808	versicolor
7	(6.7, 3.1, 4.4, 1.4)	$0.09 + 0.01 + 0.09 + 0 = 0.19$	0.436	versicolor

Step: pick the 3 nearest

- Sorted: $d_7 = 0.436 < d_5 = 0.949 < d_4 = 1.628 < d_6 = 1.808 < d_1 = 4.004 \dots$
- The 3 nearest are **Id 7, Id 5, Id 4** $\rightarrow$ versicolor, versicolor, versicolor.

Final answer**versicolor** (unanimous, 3 of 3).

- Note all 3 setosa tuples sit at distance > 4 : they have tiny petals, so they are nowhere near the query. The vote was never in doubt.

M5. Confusion matrix metrics (8 papers — highest-yield numerical in the syllabus)

DO THIS FIRST, EVERY TIME. Copy the row and column headings onto your answer sheet and mark which axis is **actual**. The IOE papers **flip the orientation between years**.

Proof: 75 Ash and 72 Ch print the *identical* four numbers 21, 6, 7, 41 — but 75 Ash has actual on the rows and 72 Ch has predicted on the rows. The result is that **sensitivity and precision swap places** (0.778 / 0.750 becomes 0.750 / 0.778). Same numbers, different answers. **verified/added**

Method

- Identify TP , FN , FP , TN from the orientation. Then $P = TP + FN$ (all actual positives), $N = FP + TN$ (all actual negatives).
- Accuracy = $\frac{TP+TN}{P+N}$ Error = 1–accuracy
- Sensitivity = recall = $TPR = \frac{TP}{P}$ Specificity = $TNR = \frac{TN}{N}$
- FPR = false alarm rate = $\frac{FP}{N} = 1 - \text{specificity}$ Precision = $\frac{TP}{TP+FP}$

Problem 5.1 5

74 Ch Rows = **Actual**, columns = Predicted. 142 / 40 / 98 / 720.

- $TP = 142$, $FN = 40$, $FP = 98$, $TN = 720 \rightarrow P = 182$, $N = 818$, total = 1000
- Accuracy = $\frac{142+720}{1000} = \mathbf{0.862}$
- Sensitivity = $\frac{142}{182} = \mathbf{0.780}$
- Precision = $\frac{142}{142+98} = \frac{142}{240} = \mathbf{0.592}$
- Specificity = $\frac{720}{818} = \mathbf{0.880}$

Problem 5.2 4

78 Bh **Same four numbers as 5.1, but rows = Predicted**. Calculate Accuracy, TPR, FPR, Precision.

- Rows are predicted $\rightarrow TP = 142$, $FP = 40$, $FN = 98$, $TN = 720$
- $P = 142 + 98 = 240$, $N = 40 + 720 = 760$
- Accuracy = $\frac{142+720}{1000} = \mathbf{0.862}$ (unchanged — the diagonal is the same)
- $TPR = \frac{142}{240} = \mathbf{0.592}$ $FPR = \frac{40}{760} = \mathbf{0.053}$
- Precision = $\frac{142}{142+40} = \frac{142}{182} = \mathbf{0.780}$
- **Compare with 5.1:** sensitivity and precision have exchanged values. This is the orientation trap, live.

Problems 5.3 and 5.4 10

75 Ash (rows = Actual) and 72 Ch (rows = predicted). Both print 21 / 6 / 7 / 41.

75 Ash — rows are Actual

- $TP = 21$, $FN = 6$, $FP = 7$, $TN = 41$; $P = 27$, $N = 48$, total 75
- Accuracy = $\frac{62}{75} = \mathbf{0.827}$
- Sensitivity = $\frac{21}{27} = \mathbf{0.778}$
- Specificity = $\frac{41}{48} = \mathbf{0.854}$
- Precision = $\frac{21}{28} = \mathbf{0.750}$

72 Ch — rows are Predicted

- $TP = 21$, $FP = 6$, $FN = 7$, $TN = 41$; $P = 28$, $N = 47$, total 75
- Accuracy = $\frac{62}{75} = \mathbf{0.827}$
- Sensitivity = recall = $\frac{21}{28} = \mathbf{0.750}$
- Specificity = $\frac{41}{47} = \mathbf{0.872}$
- Precision = $\frac{21}{27} = \mathbf{0.778}$

Problem 5.5 10

74 Ash Rows = Actual. 25 / 9 / 4 / 31.

- $TP = 25$, $FN = 9$, $FP = 4$, $TN = 31$; $P = 34$, $N = 35$, total 69
- Accuracy = $\frac{56}{69} = \mathbf{0.812}$ Sensitivity = $\frac{25}{34} = \mathbf{0.735}$
- Specificity = $\frac{31}{35} = \mathbf{0.886}$ Precision = $\frac{25}{29} = \mathbf{0.862}$

Problem 5.6 3+5

80 Ba Rows = Predicted, columns = Actual. 1050 / 250 / 150 / 950. Calculate classification error, sensitivity, false alarm rate, specificity.

- $TP = 1050, FP = 250, FN = 150, TN = 950; P = 1200, N = 1200, \text{total } 2400$
- Classification error = $\frac{250+150}{2400} = \mathbf{0.167}$
- Sensitivity = $\frac{1050}{1200} = \mathbf{0.875}$
- False alarm rate (FPR) = $\frac{250}{1200} = \mathbf{0.208}$
- Specificity = $\frac{950}{1200} = \mathbf{0.792}$ (check: $0.208 + 0.792 = 1 \checkmark$)

Problem 5.7 3+6

73 Shr 2×2 table with A = Yes, B = No. 20 / 5 / 10 / 40. Calculate TPR, FPR, precision.

- $TP = 20, FN = 5, FP = 10, TN = 40; P = 25, N = 50$
- $TPR = \frac{20}{25} = \mathbf{0.80}$ $FPR = \frac{10}{50} = \mathbf{0.20}$
- Precision = $\frac{20}{30} = \mathbf{0.667}$
- The ROC point for this classifier is (0.20, 0.80), comfortably above the diagonal.

Problem 5.8 10

70 Ch Build the confusion matrix from sequences. Actual A A B B B A A A B B; X predicts the same; Y predicts A A A A B B A A A B.

Step 1: tabulate Y position by position

	Position	1	2	3	4	5	6	7	8	9
(take A as the positive class)	Actual	A	A	B	B	B	A	A	A	B
	Y	A	A	A	A	B	B	A	A	A
	Category	TP	TP	FP	FP	TN	FN	TP	TP	FP

Classifier X

(identical to actual)

- $TP = 5, TN = 5, FP = 0, FN = 0$
- Accuracy = **1.0** Precision = **1.0**
- TPR = **1.0** FPR = **0**
- **Perfect classifier** – the ROC point is the top-left corner (0, 1).
- **Reading:** Y has decent recall (0.80) but poor precision (0.571) – it over-predicts A, firing 3 false alarms out of 5 actual Bs. Its ROC point (0.60, 0.80) is only just above the diagonal, so it is barely better than guessing.

Classifier Y

- $TP = 4, FN = 1, FP = 3, TN = 2; P = 5, N = 5$ (total 10 $\checkmark$)
- Accuracy = $\frac{4+2}{10} = \mathbf{0.60}$
- Precision = $\frac{4}{7} = \mathbf{0.571}$
- TPR = $\frac{4}{5} = \mathbf{0.80}$
- FPR = $\frac{3}{5} = \mathbf{0.60}$

M6. ANN gradients (backpropagation)**Method**

- Sigmoid and its derivative: $\sigma(z) = \frac{1}{1+e^{-z}}, \quad \sigma'(z) = \sigma(z)(1 - \sigma(z))$.
- Loss $E = \frac{1}{2}(o - t)^2 \rightarrow \frac{\partial E}{\partial o} = (o - t)$.
- Apply the **chain rule** backwards from the loss to the weight in question.

Problem 6.1 2+3+4

76 Ch 3 inputs $x_1, x_2, x_3 \rightarrow 2$ hidden nodes $h_1, h_2 \rightarrow$ output o . Weight $x_i \rightarrow h_j$ is w_{ji} ; weight $h_i \rightarrow o$ is u_i . Sigmoid activations, squared-error loss.

(a) Activation at h_1

$$h_1 = \sigma(w_{11}x_1 + w_{12}x_2 + w_{13}x_3) = \frac{1}{1 + \exp(-(w_{11}x_1 + w_{12}x_2 + w_{13}x_3))}$$

(b) Gradient of the loss w.r.t. the output o

$$E(o, t) = \frac{1}{2}(o - t)^2 \Rightarrow \frac{\partial E}{\partial o} = (o - t)$$

(c) Gradient of the loss w.r.t. w_{12}

- w_{12} is the weight from x_2 into h_1 , so the chain runs $E \rightarrow o \rightarrow h_1 \rightarrow w_{12}$.

- Write $net_o = u_1 h_1 + u_2 h_2$ and $net_{h_1} = w_{11} x_1 + w_{12} x_2 + w_{13} x_3$.

$$\frac{\partial E}{\partial w_{12}} = \underbrace{\frac{\partial E}{\partial o}}_{(o-t)} \cdot \underbrace{\frac{\partial o}{\partial net_o}}_{o(1-o)} \cdot \underbrace{\frac{\partial net_o}{\partial h_1}}_{u_1} \cdot \underbrace{\frac{\partial h_1}{\partial net_{h_1}}}_{h_1(1-h_1)} \cdot \underbrace{\frac{\partial net_{h_1}}{\partial w_{12}}}_{x_2}$$

$$\frac{\partial E}{\partial w_{12}} = (o - t) o(1 - o) u_1 h_1(1 - h_1) x_2$$

- **Sanity checks:** the factor x_2 must appear (no input, no gradient); u_1 appears and u_2 does not, because w_{12} only influences o through h_1 .
- **Weight update:** $w_{12} \leftarrow w_{12} - l \frac{\partial E}{\partial w_{12}}$.

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

Short	Means	Short	Means
DM	data mining	DW	data warehouse
DB	database	DBMS	database management system
KDD	knowledge discovery in databases	DSS	decision support system
OLTP	on-line transaction processing	OLAP	on-line analytical processing
ROLAP	relational OLAP	MOLAP	multidimensional OLAP
HOLAP	hybrid OLAP	ETL	extract, transform, load
BI	business intelligence	ER	entity-relationship
attr(s)	attribute(s)	dim(s)	dimension(s)
multidim	multidimensional	txn(s)	transaction(s)
info	information	ops	operations
repo	repository	freq	frequent / frequency
config	configuration	no.	number
PCA	principal component analysis	SVD	singular value decomposition
SMC	simple matching coefficient	KNN	k -nearest neighbour
ANN	artificial neural network	POS	point of sale
FK	foreign key	UI	user interface
GIGO	garbage in, garbage out	w/, w/o	with, without
rm	remove	vs	versus
Eg	example / for example	+Fig	draw a figure
→	leads to / therefore	+	and

Exam-paper month codes (used in the year tags): Ba = Baishakh · Jth = Jestha · Asa = Ashar · Shr = Shrawan · Bh = Bhadra · Ash = Ashwin · Ka = Kartik · Mng = Mangsir · Po = Poush · Ma = Magh · Ch = Chaitra.

Tier chips: **TOP /17** = topic asked in 7 or more of the 17 papers · **HOT /17** = 4–6 papers · **PIN /17** = 1–3 papers but worth ≥ 8 marks. Mark chips such as **2+5** are the marks that question actually carried. **verified/added** marks anything researched and added that was not in the source notes.